

The Mathematics of object-position

Calculating the Exact Render Offset Vector in the Compositing Phase

DOM Container (W_c, H_c)



$W_c = 300\text{px}$

Scaled Payload (W_p, H_p)



$W_p = 500\text{px}$

The Abstract Translation Formula

$$X_{\text{offset}} = (W_c - W_p) \times \text{PositionX}_{\%}$$
$$Y_{\text{offset}} = (H_c - H_p) \times \text{PositionY}_{\%}$$

Example Scenario: object-position: 80% 50%

If a 300px wide HTML container houses a 500px wide scaled raster image, the layout engine executes this translation:

$$X_{\text{offset}} = (300\text{px} - 500\text{px}) \times 0.80 = -200\text{px} \times 0.80 = -160\text{px}$$

Architectural Insight: The browser physically shifts the image layer 160 pixels to the left relative to the container's top-left origin (0,0). This mathematical abstraction allows the DOM layout to dictate a rigid bounding box, while the CSS rendering engine dynamically translates the uncoupled pixel payload across the sub-pixel composition plane.

Visualizing the Severed Translation Plane

